KEDGE dart v29: engineering report

What this covers: how the dart behaves with the launcher, in flight, on landing, in the soil and with the reel; which design versions were tried and why v29 was chosen; what the firmware and the rover client were tested against. All numbers come from the exact print geometry (scripts in source/). Assumptions are listed where they are used; the ones marked MEASURE should be replaced with bench numbers.

1. Summary

Item	Result	What it means
Length, mass	152 mm body (167 folded incl. augers), 262 g (111 printed + 150 bought)	Matches the 245 g beer-bottle target
Collision sweep	clean	kick 0 to 201.75° (5° steps), drills 0 to 42.25° (1° steps), every sub-part, exact meshes
Flight stability	tumbles alone; stable with 15 g nose ballast or a taut tether (simulated)	Add 15 g in the nose; reel brake 1 N or less during launch (section 3)
Launch	15.8 to 18.2 m/s for 15 to 20 m; 63 to 84 g over a 0.2 m stroke; 43.4 J	Launcher spring sizing input
Landing	301 g if it stops in 3 cm of soil	All parts checked at this load (section 4)
Holding force	dry sand 7.0 N, moist loam 44.9 N, firm clay 112.5 N	The reel must not pull harder than this: limit the wiper current
Augers in the ground	drills 43.4 mm, foot 35.5 mm	Holding rises with depth squared in sand
Battery	2.3 h with WiFi + video on; an anchor sequence uses about 6.6 mAh	Plenty

2. Design versions tried

Each version went through the full pipeline (generate, solve the leg angles from the ground, exact-mesh sweep, physics). Belly rule: nothing outside the Ø42 cylinder below the centreline.

Version	Change	Clean	Folded len	Drill depth	CG x	Margin cal	Hold loam N	Hold sand N
v29a_base	{}	yes	154.7	29.5	84.5	-0.11	31.6	5.0
v29b_long_auger	{AUGER_L: 66.0}	yes	162.7	33.9	84.8	-0.1	32.0	6.0
v29c_cells_fwd	{LAYOUT: cells_fwd}	yes	154.7	29.5	81.5	-0.04	31.6	5.0
v29d_low_pivot	{DRILL_PIV_Z: 5.0, AUGER_L: 62.0}	NO: drill_leg_L, drill_leg_L_auger	158.7	27.9	84.7	-0.11	30.2	5.5
v29e_fwd_long	{LAYOUT: cells_fwd, AUGER_L: 66.0}	yes	162.7	33.9	81.8	-0.03	32.0	6.0
v29f_fwd_long_fat	{LAYOUT: cells_fwd, AUGER_L: 66.0, AUGER_D: 16.0, DRILL_PIV_Z: 8.0}	yes	162.7	36.3	82.3	-0.03	41.3	6.6
v29g_fwd_xlong	{LAYOUT: cells_fwd, AUGER_L: 72.0}	yes	168.7	37.4	82.0	-0.02	32.4	6.9
v29h_fwd_long_foot50	{LAYOUT: cells_fwd, AUGER_L: 66.0, FOOT_AUGER_L: 50.0}	yes	162.7	33.9	81.9	-0.02	35.2	6.3
v29i_piv10	{LAYOUT: cells_fwd, AUGER_L: 66.0, AUGER_D: 16.0, DRILL_PIV_Z: 10.0}	yes	162.7	40.7	82.3	-0.03	40.1	6.4
v29j_piv12	{LAYOUT: cells_fwd, AUGER_L: 66.0, AUGER_D: 16.0, DRILL_PIV_Z: 12.0}	yes	162.7	42.4	82.2	-0.03	37.5	6.0
v29k_piv14	failed to build							
v29l_piv12_d18	{LAYOUT: cells_fwd, AUGER_L: 66.0, AUGER_D: 18.0, DRILL_PIV_Z: 12.0}	NO: drill_leg_L_auger | hull_top, drill_leg_L_auger | tail_cap	162.7	42.4	83.0	-0.04	35.1	6.7
v29m_piv11	{LAYOUT: cells_fwd, AUGER_L: 66.0, AUGER_D: 16.0, DRILL_PIV_Z: 11.0}	yes	162.7	42.8	82.3	-0.03	39.3	6.2
v29n_piv12	{LAYOUT: cells_fwd, AUGER_L: 66.0, AUGER_D: 16.0, DRILL_PIV_Z: 12.0}	yes	162.7	45.0	82.2	-0.03	38.5	6.1

Version	Change	Clean	Folded len	Drill depth	CG x	Margin cal	Hold loam N	Hold sand N
v29o_piv10_l70	{LAYOUT: cells_fwd, AUGER_L: 70.0, AUGER_D: 16.0, DRILL_PIV_Z: 10.0}	yes	166.7	43.4	82.5	-0.03	44.9	7.0
v29p_foot30	{FOOT_AUGER_L: 30.0, FOOT_TIP: 8.0}	yes	162.7	42.8	82.0	-0.05	37.5	6.0
v29q_piv8_l70	{AUGER_L: 70.0, DRILL_PIV_Z: 8.0}	yes	166.7	39.0	82.5	-0.03	41.7	7.3
v29r_piv9_l70	{AUGER_L: 70.0, DRILL_PIV_Z: 9.0}	yes	166.7	41.1	82.5	-0.02	43.2	7.1
v29s_piv10_l70	{AUGER_L: 70.0, DRILL_PIV_Z: 10.0}	yes	166.7	43.4	82.5	-0.03	44.9	7.0
v29t_piv10_l74	{AUGER_L: 74.0, DRILL_PIV_Z: 10.0}	yes	170.7	46.1	82.7	-0.02	45.3	7.6
v29u_light	{LIGHT: 1}	yes	166.7	43.4	82.4	-0.02	44.9	7.0
v29v_d17	{AUGER_D: 17.0, DRILL_PIV_Y: 31.0}	yes	166.7	43.4	82.8	-0.03	45.2	7.3
v29w_d17_l72	{AUGER_D: 17.0, DRILL_PIV_Y: 31.0, AUGER_L: 72.0}	yes	168.7	44.8	83.0	-0.02	48.5	7.7
v29x_pitch9	{AUGER_PITCH: 9.0}	yes	166.7	43.4	82.9	-0.03	45.0	7.0
v29y_l72	{AUGER_L: 72.0}	yes	168.7	44.8	82.6	-0.02	45.1	7.3

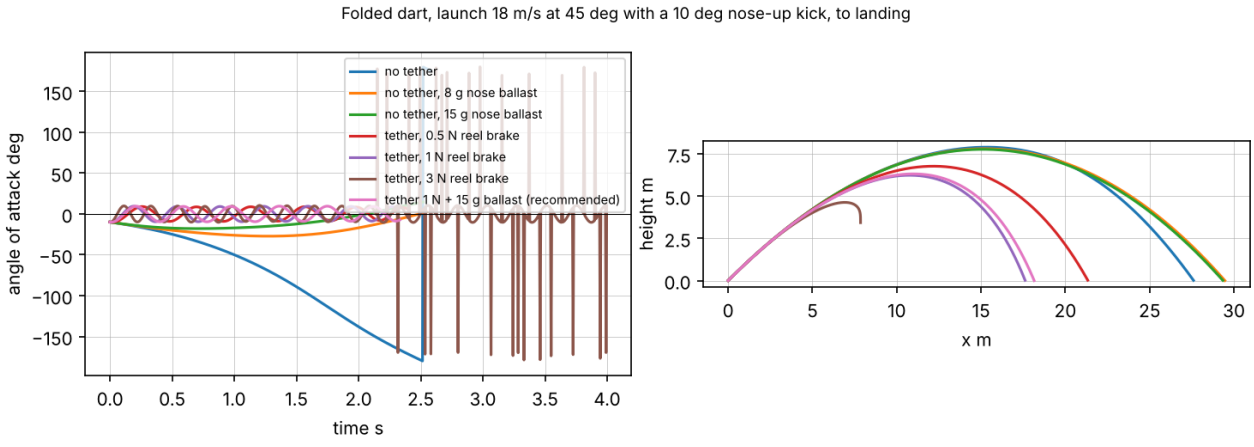
Holding columns use the final (corrected) soil model for every version; versions with pods at y 29 were swept before the 1.5 mm pod move, the final v29 sits at y 30.5.

Chosen: v29 = cells in front of the boards, Ø16 x 70 augers, drill pivot raised to z 10, pods at y 30.5 (at y 29 the Ø16 augers passed within 0.1 mm of the hull while swinging; the clearance check caught it). Final: drills deploy to 42.25° and reach 43.4 mm deep (v28: 29.5); holding in loam 44.9 N against 31.6 N for the v28-style baseline. What the study showed: raising the drill pivot lets the pods swing further before their noses touch the ground, so the augers go deeper; lowering it does the opposite and breaks the belly rule. But depth alone is not holding: with the pull horizontal, a steeper auger takes more of it sideways. Embedded length matters most, so 70 mm augers at pivot 10 beat 66 mm at pivot 11 or 12. 74 mm adds 1% for 4 mm more length past the tail; Ø17 x 72 (pods out to y 31) adds 8% for 2 mm more length: the best gain left, kept as an option; a 9 mm pitch changes nothing in this model; a lightened tray and kick keel saves under 1 g and weakens the pole; pivot 14 does not fit the deploy worm; Ø18 hits the hull folded. The 30 mm foot option (10 mm lower for the barrel) costs about 2 N; the 50 mm foot (+9 mm depth) adds about 3 N.

3. Flight

Centre of mass at x = 82.5 mm from the nose (of 152); the projected-area centre of pressure is at x = 82.0 (side) / 84.5 (top). On the body alone the dart is neutral, so it was simulated: a pitch-plane rigid-body flight with the real mass, inertia and side profile (cross-flow drag along the body), axial drag, gravity and the tether tension at the tail, launched at 18 m/s and 45° with a 10° nose-up disturbance.

Case	Max angle of attack after 0.6 s	Tumbled	Range m
no tether	179.9	YES	27.6
no tether, 8 g nose ballast	27.2	no	29.5
no tether, 15 g nose ballast	17.9	no	29.3
tether, 0.5 N reel brake	9.1	no	21.3
tether, 1 N reel brake	9.5	no	17.6
tether, 3 N reel brake	180.0	YES	7.9
tether 1 N + 15 g ballast (recommended)	9.7	no	18.2



Result: on its own the dart tumbles. Either fix works on its own; together they are robust: 15 g of ballast in the nose cavity (steel shot or nuts set in epoxy around the camera tube) and a reel that pays out with a light drag of about 1 N or less. A 3 N reel brake stalls the throw at 8 m and the dart flips at the end. So the reel must free-spool during launch with only a light brake; that matches the plan of spinning the storage reel up for launch. With ballast and a 1 N brake the throw reaches about 18 m at 18 m/s. Ballast adds 15 g (dart ~277 g) and moves the CG to about x = 78.

4. Launch and landing loads

Load case	Value	Unit	Check
cells pushed aft at launch (each)	18.2	N	rear contacts are RIGID plates on the back disc of the cell bay and the springs sit on the front disc, so launch presses the cells onto the plates and contact never breaks
cells forward at landing (both onto the front cell-bay disc)	130.0	N	that disc is 3 mm thick and carries the load through the spine to the 4 disc-1 screws
coupler bending at landing (pod 24 g, 300 g)	18.6	MPa	PLA ~45 MPa: margin 2.4
kick leg along its axis at landing (25 g)	73.0	N	taken by the Ø3 steel shaft in shear + ear bearing; worm sees almost no torque because the line of action runs through the pivot
tray + contents on 4 disc-1 screws at landing (95 g)	70.0	N per screw	M3 self-tapped into PLA shear ~250-400 N each
launcher push through the tail cap	217.0	N	4 radial M3 into the spigot ring + the hull wall in compression

Battery contacts follow from this: launch throws the cells aft, so the aft contacts are rigid plates and the springs sit on the forward disc. At landing the cells move forward into the springs for a few milliseconds; the firmware treats a reset as harmless (motors start off, WiFi is back in about a second). A 470 to 1000 µF capacitor across the 5 V output of the MP1584 rides through contact bounce and is recommended.

5. Anchoring, tipping and the reel

Soil	Each drill auger N	Foot auger N	Friction N	Total N
dry sand	2.6	0.5	1.3	7.0
moist loam	20.8	2.1	1.3	44.9
firm clay	53.2	4.8	1.3	112.5

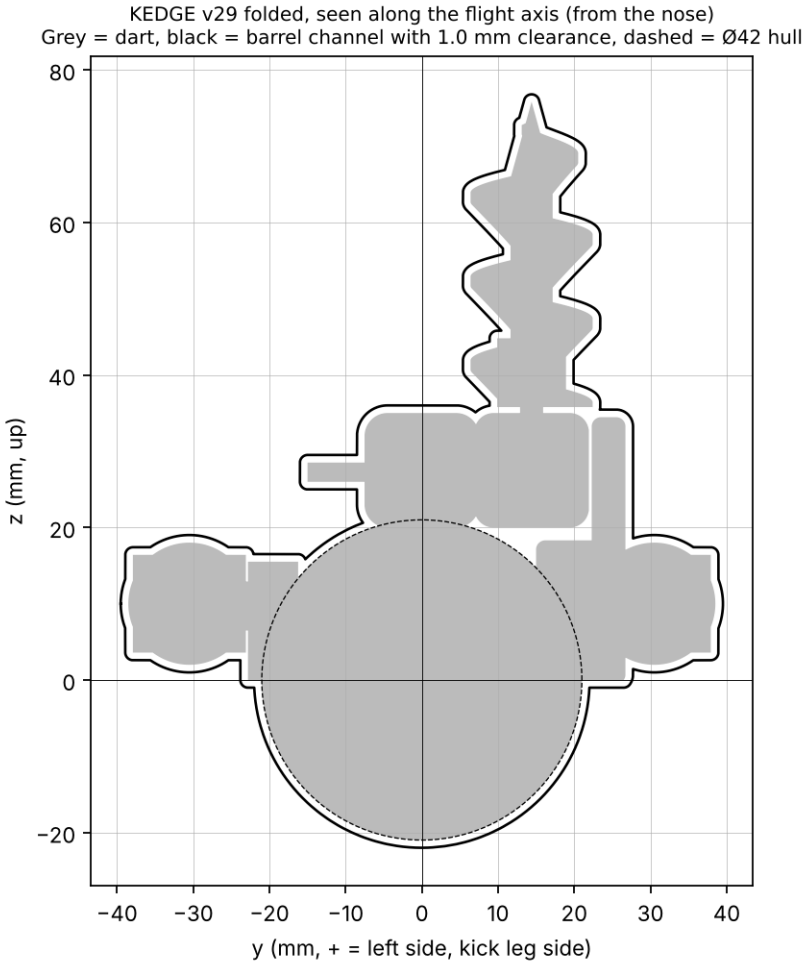
Method: each auger is a short rigid pile (Broms): sand $0.5\,\gamma\,D\,L^2\,K_p$; clay free-head solution. The drill augers are raked at 42.25°, so their embedded length is 64.6 mm for 43.4 mm of depth. The horizontal pull splits into a push along each auger (end bearing on the flights) and a sideways load; the weaker of the two limits governs. These are small anchors: length in the soil is everything. In loose dry sand the dart holds only a few newtons whatever the design; in loam or clay it holds tens of newtons. The foot's job is holding the nose down: it can resist about 14.0 N of uplift in loam.
Tipping: the tether pulls horizontally at the axis, 21 mm above the ground. At 44.9 N the nose wants to lift with 0.943 N·m; the kick foot 154 mm ahead of the drills needs to hold 6.14 N down. That is why the foot is planted first and the kick worm stays locked.
Reel interface: the reel's wiper motor can pull far more than any of these numbers. Limit tether tension on the reel side: read the BTS7960 current (IS pins) or add a load cell (you have the 100 kg cell + HX711) and stop or slip when tension passes the soil's holding value (firmware/reel/reel_tension_limit does this: speed tapers over the last 30% and stops at the limit; set it with l20). Start with 20 N, raise it while watching the dart's pitch on /status: a nose-up change of more than about 5° means it is starting to drag.

6. Motors, battery, strength

Kick: the leg family weighs 24.6 g with its centre 76.7 mm from the pivot, so gravity asks at most 0.189 kg·cm of the worm. Small worm N20s give 1 to 4 kg·cm: ample. Deploy: both pods together need 0.157 kg·cm against gravity; the real load is soil, which is why the drills spin while the pods swing (firmware auto sequence).
Important geometry fact: the drill augers sit radially on the pods, so their depth comes only from the swing. They cut sideways through the soil as the pods rotate; spinning them while deploying is what lets the deploy worm push them in. The foot auger is different: near 180° its axis is almost vertical and the swing drives it along its own axis, so it really screws in.
Battery: 5.18 Wh. WiFi + VTX + camera draw about 2.22 W: 2.3 h. Worst case all motors stalled is about 2 A (BMS 8 A, INA219 trip set at 2.5 A).
Coupler torque capacity about 1.01 N·m (PLA, derated for the flat and layer lines): well above the deploy worm. Kick pole (round pole + keel, section modulus 94.3 mm³ from the real cross-section): 15.3 MPa at a 20 N sideways load on the foot; keep sideways loads under about 29.5 N (factor 2 on PLA). The pole prints lying down, so the bending runs along the extrusion lines.

7. Launcher interface

The barrel must clear this silhouette (folded dart seen along its axis). Channel with 1.0 mm clearance spans y -39.5 to 39.5, z -22.0 to 76.8 mm. Files: barrel_profile.dxf (sketch it in SolidWorks or Fusion and extrude), barrel_keepout.stl (subtract it from the barrel body), barrel_profile.svg, and KEDGE_v29_assembly_folded_STEP.zip (the whole folded dart as one STEP assembly to design the barrel around).



the belly (z < 0) is the plain Ø42 cylinder over x 20..141: cradle it there; the barrel's lower half can be a plain Ø44 half-pipe. push on the tail cap end face (flat ring Ø42, tether holes in the middle): keep the pusher face a ring so the tether exits through its centre.

The folded foot auger is the tallest thing on the dart (top of the silhouette). If the barrel slot for it is a problem, the options in order of cost are: a shorter foot auger (30 mm: about 10 mm less height, about 8 mm less depth), or a foot auger tilted 45° toward the tail in the housing (folded it leans back over the hull; deployed it points down and forward, which resists the tether better). Neither is in v29; both are quick to generate on request.

Spring energy needed: about 43.4 J delivered to the dart for 20 m, more with losses. A longer stroke lowers the peak load on the dart: 0.4 m halves the g.

8. Firmware and rover link

The control logic is a plain C++ header (kedge_logic.h) shared by the sketch and a PC test that drives it against a motor model. The auto sequence is: kick swings out; when the kick current rises (foot auger touches) the foot auger starts; the kick keeps pushing until the housing lands, read by pausing the foot for 150 ms every 400 ms so the INA219 sees the kick alone; the foot finishes; then the pods deploy with the drills spinning, deploy stall read the same way with the drills paused; drills finish; hold.

Rover API on the KEDGE WiFi: GET /status (JSON: rail volts, current, peak, motors, sequence, armed, landed, roll, pitch, last event), GET /key?c=&ms;=, GET /stop, GET /auto. kedge_client.py (standard library only) gives status, watch-to-CSV, hold-a-key, auto with progress, terminal driving, and an FPV window with the telemetry over the RC832 + capture card picture (OpenCV).

Landing detection: the BNO085 accelerometer sees the launch spike, flight, impact and stillness; with A (arm) pressed before launch the dart anchors itself 1.5 s after it comes to rest.

Test results (PC simulation of the real logic, and the client against a mock dart):

0 ms auto: kick swing

3150 ms auto: foot touches, foot auger on

5310 ms auto: kick down, foot finishing

6810 ms auto: deploy with drills spinning

8970 ms auto: pods down, drills finishing

11970 ms auto: anchored, holding (worms self-lock)
ok: auto sequence reaches 'anchored'
ok: foot auger starts when the foot meets the ground (kick current rise)
ok: kick stops after the housing lands (stall read with the foot paused)
ok: deploy starts only after the foot is planted
ok: deploy stops on its own stall even though the drills draw more current
ok: drills finish after the pods stop
ok: everything off at the end
ok: manual keys are ignored while auto runs
ok: space aborts the auto sequence and stops everything
ok: over-current trip stops the motors
ok: WiFi hold: repeats keep the kick running
ok: WiFi hold: key released -> stops within the 350 ms deadman
ok: serial key runs 1.5 s
ok: serial key stops after 1.5 s
ok: kick never meets the ground -> aborts at the time limit, nothing left running
ok: landing detector: launch spike, flight, impact, stillness -> landed
ok: landing detector: bench handling alone never triggers
ok: landing detector: soft landing with the impact spike missed still triggers
ok: landing detector: reset re-arms it (A pressed again)
status: {"t_ms":1234,"rail_V":7.60,"current_mA":0,"peak_mA":0,"speed":255,"motors":{"deploy":0,"kick":0,"drills":0,"foot":0},"seq":"idle","armed":false,"landed":false,"roll":1.5,"pitch":-3.2,"event":"boot"}
ok: status JSON
ALL PASS
ok: status
ok: hold sends 7 repeats, max gap 152 ms (< 350 ms deadman), then stop
0.0s kick swing 120 mA auto: kick swing
1.0s kick + foot planting 120 mA auto: kick + foot planting
1.6s foot finishing 120 mA auto: foot finishing
2.0s deploy + drills 120 mA auto: deploy + drills
2.9s drills finishing 120 mA auto: drills finishing
3.5s anchored 120 mA auto: anchored
ok: auto followed to anchored
ALL PASS
Compiled for XIAO_ESP32S3 on ESP32 Arduino core 2.0.17 and 3.3.12 (USB CDC On Boot: Enabled; the sketch refuses to compile with it disabled, because Serial would then take D6/D7).
Independent reviews: two separate code reviews of the firmware and wiring and one of the geometry checks and physics were run by agents that had not seen the work being produced. They found, among others: 3V3 must go to the DRV8833 SLP pad, never VCC/VIN; D6/D7 toggle at reset (drills moved to D8/D9); safety logic must not wait on the web server (now its own 5 ms task); Shift-release and touch-scroll could leave a motor running on the web page; the clay holding and the pole strength formulas were wrong; the sweep step was too coarse for the leg tip. All were fixed and re-checked.

9. Open items to measure

• Weigh every bought part on the kitchen scale and update BOUGHT_G in physics.py (the mass and CG use catalogue values).
• Bench stall currents of each motor at 8 V: set kickContactMa / kickStallMa / deployStallMa in kedge_logic.h.
• Launcher stroke and spring: replace STROKE_M and re-run physics.py for the real g loads.
• A pull test in your soil: plant the dart, pull the tether with a luggage scale, compare with section 5.
• BMS stick dimensions when it arrives; F693ZZ flange size (catalogue 9.5 x 0.6 used).